

CS & AI • EXPLAINED

Why do computers only understand 0s and 1s?

Every photo, song and video — secretly just two digits.

THE HOOK

Every photo, song, video and WhatsApp message on your phone is **ultimately** just 0s and 1s.

But why only **two** digits — when we humans count with **ten**?

In 5 minutes, the whole “binary” mystery disappears. 



TODAY'S QUESTION

Why does a computer use just **two** symbols —
and how can that store
everything?

Inside: billions of tiny switches.



OFF

= 0



ON

= 1

These switches are called **transistors**.

- A computer is built from **billions** of microscopic switches.
- Each one is either **OFF** or **ON** — two states.
- OFF = 0, ON = 1. That's the whole alphabet.

Two states are easy to tell apart.

- Like a light switch — clearly **on** or **off** (high or low voltage).
- Ten different levels? Easy to **confuse** → errors.
- Simple = **reliable** = fast. This is the key insight.

2 LEVELS

Hard to mistake **ON** for **OFF**.

10 LEVELS

Is that a 6 or a 7? Too risky.

IDEA 3 / 4 — LINE THEM UP TO COUNT

One switch is small. **Many** can count anything.

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one switch = 1 bit = 2 options

line up 4 bits → read like place value

8	4	2	1	
1	0	1	0	→ 8 + 2 = 10

- Each 0/1 is a **bit**.
- **8 bits = 1 byte = 256** combinations.
- Enough bits → **any** number you want.

Numbers → **binary**. So binary stores it all.



letters

→ a code (ASCII / Unicode)



colours

→ Red, Green, Blue values



sound

→ thousands of samples

- First turn it into **numbers**...
- ...then every number becomes **binary**.
- One simple system stores **everything**.

WHY IT MATTERS

This is why “digital” is so reliable.

- Two clean states is why computers **rarely make mistakes**.
- Simple switches can be made **tiny** — billions on one chip.
- Tiny + simple = **cheap and fast**. That’s the whole digital world.

COMMON MYTH — ”YEH GALAT SAMAJHTE HAIN”

X MYTH

”Binary is some weird, **secret code** that only computers understand.”

✓ REALITY

It's just **counting with two symbols** — because the hardware physically has two states: on / off switches. Nothing mysterious.

RECAP — 2 THINGS TO REMEMBER

- 1 A computer = billions of **on/off switches** — so its alphabet is just 0 and 1.
- 2 Two states are **reliable**, and combining them can represent **any** data.

0s and 1s = switches you can trust. 

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Next question: How does a computer store a photo? 